There are still a great number of unknowns about asteroid mining and many questions will remain unanswered until mining operations occur. Perhaps the mining techniques of a company may unintentionally cause significant damage to the lunar or asteroid environments. That technique could then be prohibited through an agreement written after the first missions have occurred. Similarly, certain minerals or regions of an asteroid or the Moon may need to be managed in a way that is not currently apparent.

Although this governance method seems risky, it is a smart choice for the industry. It is likely the US would strongly oppose ratifying a space mining agreement before any mission's launch, especially since the majority of space mining companies are US based. There are currently sufficient international and domestic regulations for space mining. Instead of taking years to draft a new agreement governing space mining, the space community should give space mining companies the regulatory freedom needed to get their missions off the ground. Even then, an agreement should only be considered if there is a distinct legal challenge that companies encounter. A new space activity should not automatically require a new international agreement. If an agreement is eventually deemed necessary, it needs to incorporate concepts from related industries to maximize effectiveness and minimize redundancy with previous agreements.

IV. CONCLUSION

Space mining is an entirely new outer space activity from both a technical and regulatory standpoint. We can only predict what space mining missions will entail and what level of success companies will achieve. Mission and technology development for space mining is much more challenging than anyone contemplated, as evidenced by missions that are behind schedule in development and two of the first mining companies being acquired by other companies within a few months of each other.

Sample-return missions, where a spacecraft collecting a small sample of materials from a celestial body besides the Earth, are themselves fairly low in frequency. There have been several sample-return missions to the Moon, including the Apollo lunar landing missions, but the first sample-return from an asteroid was accomplished in 2010, when the Japan Aerospace and Exploration

Agency's (JAXA) Hayabusa spacecraft returned a small amount of material to earth from the asteroid Itokawa.[72] JAXA's Hayabusa2 spacecraft and NASA's OSIRIS-REx spacecraft will return an asteroid sample to Earth in 2020 and 2023 respectively, but sample-returns are still considered a novel mission.[73] Transforming sample-return missions into mining operations with heavy equipment could take decades. This new frontier has presented quite the legal challenge, but it is time to move past debate and begin talking about management systems, since the prospect of the improper management of space mining likely drives opposition of space mining from select nations.

While The Hague Working Group's Building Blocks provide a solid foundation for discussion of the operational aspects of commercial space mining, it is too early to adopt legally binding principles on commercial space mining operators. The dialogue between multiple types of industry stakeholders on this issue should continue, but The Hague Working Group should consider implementing principles derived from those established for the deep-sea mining and telecommunications industries. The establishment of an international regulatory authority on commercial space mining, as called for by the Moon Agreement, should not be executed in the near term.

Space mining calls for a forward-thinking legal regime and not something that is derived entirely from the past. Space mining could become a trillion-dollar industry if it is managed effectively and at the appropriate time. If domestic companies are hampered by agreement negotiations and legal debates, the industry will not prosper. The international community has a duty to permit space technology innovation and the only way to do this for space mining is to delay the implementation of a binding agreement until the industry seems stable. At that point, the agreement should adopt principles from related sectors to improve the likelihood of ac-

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[72] *What is the Asteroid Explorer "Hayabusa"?* JAPAN AEROSPACE EXPLORATION AGENCY, https://global.jaxa.jp/projects/sas/muses_c/files/presskit_hayabusa_e.pdf (last visited Aug. 25, 2019).

[73] *See, e.g., Asteroid Explorer Hayabusa2*, JAPAN AEROSPACE EXPLORATION AGENCY, http://www.isas.jaxa.jp/en/missions/spacecraft/current/hayabusa2.html (last visited Aug. 25, 2019); *Mission*, UNIV. OF ARIZ., https://www.asteroidmission.org/ (last visited Aug. 25, 2019).

ceptance by space-faring nations. Governance of outer space activities is critically important, but it is just as important to make sure that the governance is well-reasoned, fair and timely. Anything less puts a new space activity at risk of collapse due to regulatory burden.

ANDY WEIR, *ARTEMIS*

Review by Christian J. Robison

Artemis, the most recent novel released by Andy Weir (author of the critically acclaimed work, *The Martian*), tells the story of Jasmine Bashara, a young woman who resides on the Moon. Weir creates a fascinating world in which "Jazz" lives and works in Artemis – a Moon colony made up of five intricate domes that mainly caters to wealthy tourists and entrepreneurs who seek to utilize the resources of the Moon or to visit points of interest on the lunar surface (such as the Apollo 11 Moon-landing site).

Unfortunately, Jazz is not often able to enjoy the wonders of Artemis, as she is a meagerly paid porter living in the most austere conditions Artemis can offer, forced to smuggle prohibited goods into the colony to make ends meet. Until, one day, Jazz is presented an opportunity to use her more nefarious skills to change her fortunes. Weir's exciting, suspenseful and page-turning plot recounts how Jazz's decision to engage in a get-rich-quick scheme quickly embroils her in a criminal conspiracy that threatens the very future of Artemis itself.

With *The Martian*, Andy Weir demonstrated his ability to conjure a "realistic" science fiction novel in which the protagonist utilizes existing scientific knowledge and technology to survive the dangers presented by space exploration. *Artemis* is no different. Weir goes to great lengths to explain the inner-workings of Artemis and the Moon environment in easy-to-understand terms that can be enjoyed by many audiences. The author is diligent in explaining how the Moon base operates on a daily basis and how Jazz exploits the scientific principles of both Artemis and the Moon to ultimately save the day.

Though many scientists and researchers may be happy with how Weir has carefully constructed the setting and plot of *Artemis*, legal scholars may not be so pleased with Weir's characterization of international space law. In *Artemis*, Weir simply states that "maritime law" governs outer space. This echoes Weir's similar assertion in *The Martian*, where the author even drafts a narrative in which

the protagonist, Mark Watney (later famously portrayed by Matt Damon in the film adaptation), deems himself a "space pirate" because of this legal application. Many of the principles that make up maritime law, such as foundational principles concerning sovereignty and resource exploitation, also make up the foundations of international space law. Nonetheless, most, if not all, legal scholars characterize the law of outer space as its own distinct legal regime.

Despite Weir's mischaracterization of international space law,[1] Weir still may have created a world in which Artemis and its daily activities exist in line with principles of international space law, chiefly the non-appropriation principle. In fact, *Artemis* may incidentally provide an initial blueprint for how a future human community on a different planet or other celestial body could legally operate.

The non-appropriation principle as espoused by Article II of the Outer Space Treaty[2] has been the subject of vast legal scholarship for quite some time. It has been interpreted not only by legal scholars, but also by individual States through the passage of domestic legislation.[3] However, notwithstanding this immense discussion, the non-appropriation principle set forth by the Outer Space Treaty essentially prohibits States and perhaps private actors from (1) claiming sovereignty over celestial bodies, and (2) pre-

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[1] Andy Weir does not make reference to the seminal Outer Space Treaty in either of his novels. Considering that both works are set in the future, one could argue that Weir's works exist in a "timeline" different from our own, in which a different legal regime was chosen by the international community to govern outer space. This scenario, however, seems unlikely considering Weir's realistic settings and plot lines which are rooted in proven scientific principles and existing technology, as well as reference real-world historical events such as the 1969 Apollo Moon-landing.

[2] Article II states, "Outer Space, including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means." Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space Including the Moon and Other Celestial Bodies art. II, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 [hereinafter Outer Space Treaty]. For a brief summation of both the accepted and debated foundations of the non-appropriate principle, see Jinyuan Su, *Legality of Unilateral Exploitation of Space Resources Under International Law*, 66(4) INT'L & COMP. L. Q. 991 (2017), *available at* https://www.cambridge.org/core/journals/international-and-comparative-law-quarterly/article/legality-of-unilateral-exploitation-of-space-resources-under-international-law/EE17641F7B7C6404A79B77AEB627D5F4/core-reader.

[3] *See, e.g.,* U.S. Commercial Space Launch Competitiveness Act, Pub. L. No. 114-90, § 401, 129 Stat. 70 (2015).

venting others from utilizing outer space resources. Again, the specific nuances of these general assertions are the subject of much scholarly debate (as demonstrated by the very articles contained in this volume of the *Journal of Space Law*), especially since the non-appropriation principle has not been fully tested against potential endeavors like a Moon base or space mining. Overall, most scholars of international space law would agree with at least the two elements concerning the non-appropriation principle elaborated above.

On Weir's Moon, the Kenya Space Corporation (KSC), a private entity headquartered in Kenya, maintains the Moon-base. KSC was able to use a terrestrial resource, the equator, to convince multiple State governments to invest millions of dollars in KSC; the portion of the equator located in Kenya would prove to be the perfect launch point for KSC to construct Artemis and to provide transportation from Earth. KSC delivers passengers and cargo from every stretch of the globe and converts world-currencies into "slugs" that can be used for transactions within Artemis. KSC also partners with Sanches Aluminum in order to provide an almost endless supply of oxygen to Artemis. Specifically, Sanches Aluminum collects and smelts the plentiful amount of aluminum found on the surface of the Moon in order to produce an oxygen by-product. In exchange for oxygen, Artemis provides Sanches Aluminum an exclusive permanent contract for free energy produced by the colony.

This operation and exchange is extremely beneficial to both entities, until Jazz learns that Sanches Aluminum is simply a money-laundering front for O Palacio – an organized crime syndicate located in Brazil. As Jazz further discovers, O Palacio can exploit its exclusive contract with Artemis in order to take advantage of new technology produced by the Moon colony. Essentially, O Palacio plans to use its criminal enterprise to extort Artemis for its own financial gain, thus transforming Artemis into a virtual mob-State.

Though a criminal plot may compromise the daily operations and overall character of the partnership between Artemis and Sanches Aluminum, this does not change the fact that the world Andy Weir creates initially falls in line with the accepted foundations of the non-appropriation principle. The Moon base itself does not claim sovereignty over any portion of the Moon. Similarly, KSC